



MAKE UP EXAMINATIONS - APRIL 2021

Program	: B.E. : Computer Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit

UNIT- I

- Write recursive and iterative versions to find out Sum of recursive functions. CO1 (10)
 - Write a C function to concatenate two strings using pointers which has to be named as strcat_user(), that accepts two string pointer variables. Do not use any string related predefined functions in your code. Your user defined C function should Concatenate two strings, then return the pointer to the destination string. Use only character pointers for accessing each character from/to string variables. CO1 (10)
- Write a C function for multiplying a Sparse Matrix by a scalar value. Store the resultant in Sparse matrix representation. The sparse matrix is represented in triplet format < row,col,value>. The sparse matrix is to be stored using array of structure where the triplet format is stored. CO1 (10)
 - Write a C function named padd() that add two polynomials. Note: There is only one array of structure to store two input polynomials and resultant polynomials. CO1 (10)

UNIT - II

- Apply infix to postfix conversion algorithm to convert following infix expressions to postfix expressions:
 - A-B/(C*D/E) * F
 - A+B*C+D*E/F/(G-H).
 - Write C functions for add () and delete () operations over a multiple queue. CO2 (08)
 - List any four applications of a queue. CO2 (04)
- Define queue. Illustrate basic operations on a linear queue with suitable diagrams. CO2 (06)
 - Write a C function to check whether a string is palindrome or not using stack. CO2 (06)
 - Write C functions to perform add and delete operations on a circular queue using array implementation. CO2 (08)

UNIT - III

- What do you understand by a linked list? Write a C function SEARCH (F, X) that accepts a pointer P to a list of integers and an integer X, and returns a pointer to a node containing X, if it exists, and the NULL pointer otherwise. CO3 (07)
 - Write a C function to insert and delete a node from the front end in case of doubly linked list. CO3 (07)
 - Write an algorithm to find whether a given list is ordered or not. CO3 (06)

6. a) Write advantages of doubly linked list over singly linked list. Write C function that will insert a given integer value into an ordered doubly linked list. CO3 (07)
- b) What is a single linked list? Explain with an example how a single linked list can be used for sorting a set of N numbers. CO3 (07)
- c) Write graphical representation to show basic operations of list. CO3 (06)

UNIT – IV

7. a) Define with an example each of the following: CO4 (06)
i) Tree ii) Degree of a tree iii) Height of a tree.
- b) Write an iterative C function to search for an element into a binary search tree. CO4 (06)
- c) Suppose that we have the following key values 7,16,49, 82, 5, 31, 6, 2 CO4 (08)
Write out the max heap after each value is inserted into the heap.
8. a) Define with an example each of the following: CO4 (06)
i) Binary tree
ii) Full binary tree
iii) Complete binary tree.
- b) Construct a binary search tree stepwise for the following list of numbers: 24, 12, 6, 18, 30, 54, 7, 10, 42, 36. Perform inorder, preorder and postorder traversal on the constructed binary search tree. CO4 (08)
- c) Write a C function to delete a value from a max heap. CO4 (06)

UNIT – V

9. a) Show multilist representation of a graph with an example. CO5 (08)
- b) Write a C function for breadth first traversal over a graph. Show the breadth first traversal of a graph with an example. CO5 (08)
- c) Relate AVL tree and balance factor. CO5 (04)
10. a) How are adjacency matrix and adjacency lists useful for a graph? CO5 (08)
- b) Define depth first traversal of a graph. Show it with an example. CO5 (06)
- c) Show the purposes of winner tree and loser trees. CO5 (06)
